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TRACKSHIELD

AN INTELLIGENT AI-IOT SYSTEM FOR RAILWAY TRACK HEALTH AND INTRUSION DETECTION

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Why TrackShield is proposed

Railway Safety Challenge

Tracks can face people or animals entering the track, objects or stones obstructing the track, and visible cracks or rail damage.

Integrated Monitoring

TrackShield combines track intrusion detection and track-condition monitoring into one AI and IoT-based system.

Detection & Processing

HuskyLens detects humans, animals and other obstructions; a close-view camera monitors the rail surface for possible cracks or defects.

Early Warning

A central controller processes detected information and provides localized warnings, displays the condition and can send an IoT-based alert.

Abstract

Project overview and motivation

TRACKSHIELD – ABSTRACT

Railway transportation requires continuous monitoring to ensure the safety of passengers, railway infrastructure, and surrounding areas.

Track damage, abnormal vibrations, and unauthorized intrusion onto railway tracks can lead to serious accidents and operational disruptions.

TrackShield combines track condition monitoring with real-time intrusion detection. An ESP32 collects data from an accelerometer and vibration sensor, while a HuskyLens AI camera detects and recognizes people or objects entering the restricted track area.

When abnormal conditions or unauthorized intrusion are detected, the system provides immediate warning through a buzzer and display and can transmit alerts through an IoT platform.

The proposed system is a low-cost prototype demonstrating continuous railway safety monitoring and early warning capabilities.

Literature Survey

Key studies and how TrackShield addresses their gaps

Study	Methodology / Focus	Limitation	TrackShield response
Ji et al. (2021)	Deep-learning approaches for railway track / obstacle detection	Needs suitable datasets and real-world deployment validation	Low-cost vision prototype combines intrusion detection with track monitoring
Ing & Li (2022)	Review of railway defect detection and inspection methods	Inspection can be time-consuming and resource-intensive	Edge/IoT architecture supports live hazard alerts and remote monitoring
De Donato et al.	Audio-video deep learning for railway scene understanding	Different datasets and sensing modalities can limit generalization	Simpler vision-first prototype with sensor-based track-condition monitoring
Nan et al. (2025)	LiDAR-camera joint obstacle detection	Higher hardware cost and integration complexity	Lower-cost vision approach; sensor fusion can be extended as needed
Zhao et al. (2024)	Lightweight CNN with improved railway detection	Embedded complexity and training-data requirements	Lightweight prototype targets relevant hazards with accessible hardware
Radosavljevic et al.	Rail defect detection and embedded monitoring	Accuracy, energy use, and hardware constraints affect deployment	Embedded/IoT-friendly architecture provides local alerts and remote updates
Moorthy & Rukkum	Vision-based wildlife intrusion detection	Lighting, background, and species variation affect detection	Animal-specific intrusion classes can be incorporated into the camera system
Guan, Ieiri & Yoshie	ROD-YOLO railway obstacle detection	Higher computation and training requirements can limit deployment	Compact low-cost prototype combines track monitoring and intrusion alerts

Base Paper & Proposed Improvement

IoT-based rail track condition monitoring and derailment prevention

BASE PAPER METHODOLOGY

Authors: C. Chellaswamy, T. S. Geetha, A. Vanathi, K. Venkatachalam

- IoT-based Railway Track Monitoring System (IoT-RMS)
- MEMS accelerometers measure vibration signals generated during train movement.
- Abnormal vibration patterns identify possible track irregularities.
- Location of detected irregularities can be determined.
- Track information is transmitted to a cloud platform for remote monitoring.
- Alerts can be shared with other trains for early warning.

LIMITATIONS IDENTIFIED

- Mainly focuses on vibration/accelerometer abnormalities
- No AI-based visual intrusion detection
- Unauthorized people/objects are not separately addressed
- Architecture is primarily vibration-based

TRACKSHIELD CONTRIBUTION

- Combines track health monitoring and intrusion detection
- Adds HuskyLens AI camera for people/objects
- Provides buzzer and display alerts
- Supports IoT-based remote monitoring

Base Paper: Experimental Data & Contributions

EXPERIMENTAL DATA

- Four different track abnormality conditions were tested.
- Experiments were performed at different train speeds.
- Field experiments were carried out on railway routes around Chennai.
- Vibration measurements were the primary sensing information for identifying abnormal track conditions.

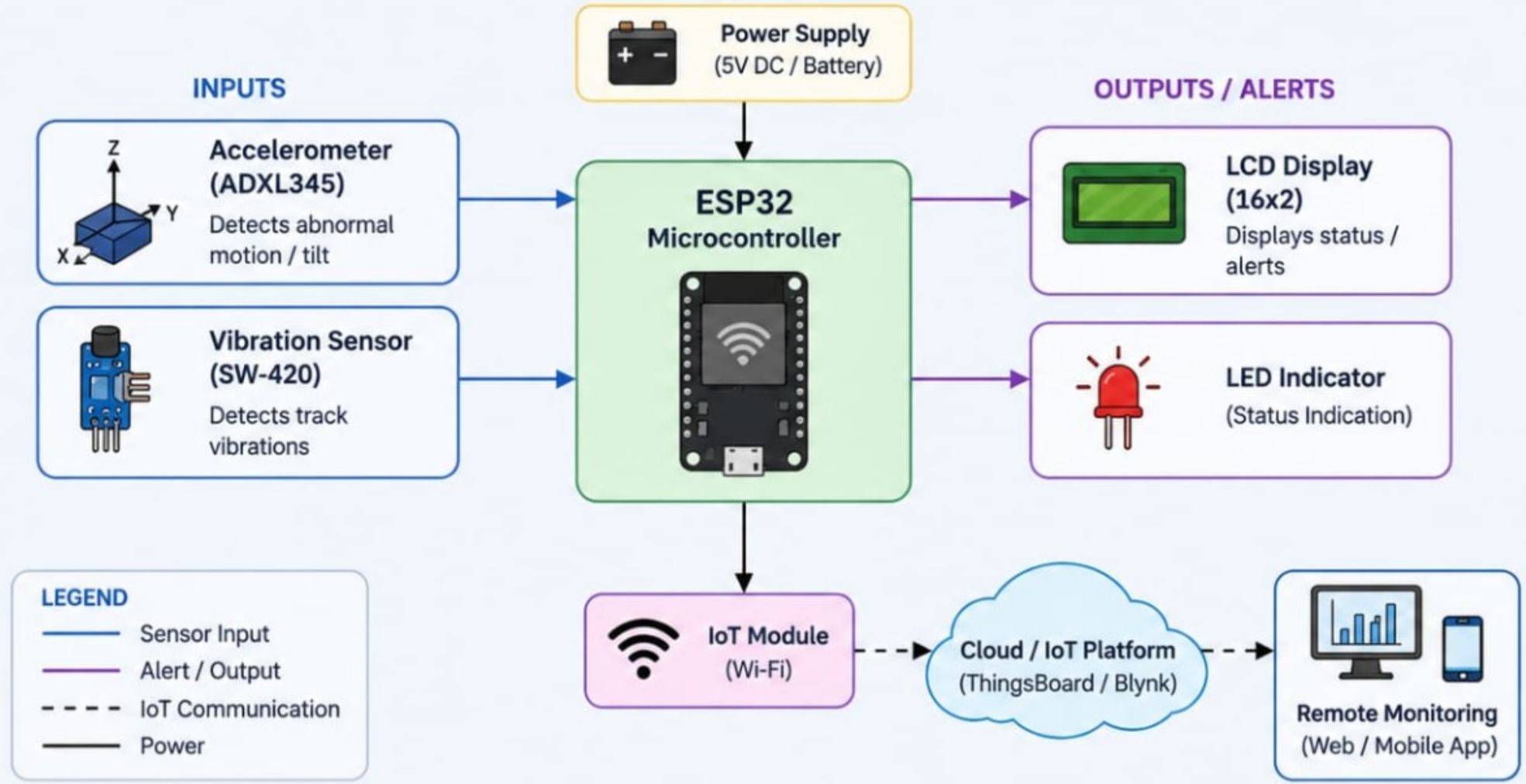
MAJOR CONTRIBUTIONS

- Early detection of railway track abnormalities
- IoT-based cloud monitoring of track condition information
- Identification of the location of track irregularities
- Early warning capability intended to help reduce derailment risk

Objective: detect abnormalities at an early stage so corrective action or warning can be provided before the abnormality contributes to a dangerous situation.

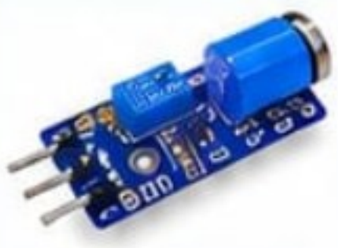
TrackShield Architecture

Sensor inputs → ESP32 → local alerts + IoT cloud + remote monitoring



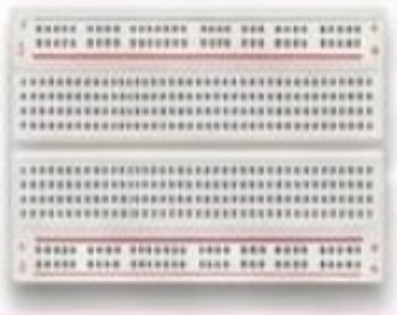
TrackShield – Required Libraries / Modules

(Part 1 : Core, Sensors & Input Modules)

No.	Module / Library	Library Name	Brief Explanation
Core Controller			
1	 **ESP32 DevKit V1** (Controller)	**ESP32** (built-in / board package)	Main controller of the project. Provides Wi-Fi, reads sensors, controls outputs and handles IoT communication.
Input Modules (Sensors)			
2	 **ADXL345** Accelerometer	**Adafruit_ADXL345** (Adafruit sensor library)	Detects motion, tilt and acceleration. Used to identify abnormal movement.
3	 **SW-420** Vibration Sensor	**No library required** (use digitalRead())	Detects vibration or shock. Gives digital HIGH/LOW output.
IoT / Network			
5	 **Wi-Fi** (ESP32 built-in)	**WiFi** (comes with ESP32)	Connects to your Wi-Fi network for cloud/remote monitoring.
6	 **Blynk** (IoT Platform)	**Blynk** (Blynk library)	Sends data to mobile/cloud. Used for remote monitoring and alerts.

TrackShield – Required Libraries / Modules

(Part 2 : Output Modules, Display, Power & Others)

No.	Module / Library	Library Name	Brief Explanation
Output Modules			
7	 16 x 2 LCD (I²C)	LiquidCrystal_I2C (LiquidCrystal library)	Displays status, alerts and messages. Uses I²C for fewer pins.
8	 Buzzer	No library required (use tone() / digitalWrite())	Provides audio alert when abnormal activity is detected.
9	 LED	No library required (use digitalWrite())	Shows system status (e.g., normal / alert).
Supporting Components			
10	 5V Battery / Power Supply	No library required	Provides 5V power to the ESP32 (or use a 5V adapter / power bank).
11	 Breadboard	No library required	Used to make connections between modules without soldering.
12	 Jumper Wires	No library required	Connects all modules and components.
13	 Resistors (for LED/buzzer)	No library required	Limits current for LED and other components (e.g., 220Ω).
Final / Cloud Platform			
14	 Cloud Platform (Blynk / ThingsBoard)	Blynk / ThingsBoard (Cloud service)	Used for remote monitoring and data logging. (Optional)

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